

DPT Exercises

Exercise sheet 3

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Exercise 1. We consider a fair tetrahedron whose faces are labeled with the numbers $\{1, 2, 3, 4\}$. Assume the tetrahedron is rolled $k \geq 1$ times. Let E be the event that the same number appears in each roll. Furthermore, let F be the event that the product of the k rolled numbers is even.

- (i) Compute the probability of E .
- (ii) Compute the probability of F .
- (iii) Compute the probability of $E \cup F$.

Solution: Let $\Omega = [4]^k = \{1, 2, 3, 4\}^k$ be the sample space consisting of all possible sequences of length k . Each outcome $\omega \in \Omega$ is a k -tuple $(x_1, x_2, \dots, x_k)$, where $x_i \in [4]$ for all $i \in [k]$. Since the tetrahedron is said to be fair, and the sample space is finite, we can apply the Laplace distribution and specify the probability of a single outcome in Ω as

$$P(\{\omega\}) = \frac{1}{|\Omega|} = \frac{1}{|[4]^k|} = \frac{1}{4^k}.$$

- (i) Let $E \subset \Omega$ denote the event that all rolls yield the same number. That is, $E = \{(i, \dots, i) \in \Omega \mid i \in [4]\}$. Each such tuple uniquely corresponds to a single number in $[4]$, so $|E| = 4$, and

$$P(E) = \frac{|E|}{|\Omega|} = \frac{4}{4^k} = \left(\frac{1}{4}\right)^{k-1}.$$

- (ii) To compute the probability of event F , we consider the complementary event F^c , where the product is odd. An odd product occurs exactly when all k rolls result in numbers from the set $\{1, 3\}$. We can express the set of outcomes corresponding to F^c as $\{(x_1, \dots, x_k) \in \Omega \mid x_i \in \{1, 3\} \ \forall i \in [k]\} =: \{1, 3\}^k$, with $|F^c| = |\{1, 3\}^k| = |\{1, 3\}|^k = 2^k$. Thus:

$$P(F) = 1 - P(F^c) = 1 - \frac{|F^c|}{|\Omega|} = 1 - \frac{2^k}{4^k} = 1 - \left(\frac{1}{2}\right)^k.$$

- (iii) We compute the probability of the event $E \cup F$ using the inclusion-exclusion formula:

$$P(E \cup F) = P(E) + P(F) - P(E \cap F).$$

For the event $E \cap F$ to occur, all k rolls must result in the same number i and yield an even product, so $i \in \{2, 4\}$. In more formal terms, $E \cap F = \{(i, \dots, i) \in \Omega \mid i \in [4]\} \cap \{(x_1, \dots, x_k) \in \Omega \mid x_i \in \{2, 4\} \ \forall i \in [k]\} = \{(i, \dots, i) \in \Omega \mid i \in \{2, 4\} \ \forall i \in [k]\}$, with $|E \cap F| = 2$ since each tuple in $E \cap F$ uniquely corresponds to a single number in $\{2, 4\}$. Therefore,

$$P(E \cap F) = \frac{|E \cap F|}{|\Omega|} = \frac{2}{4^k} = \frac{1}{2} \cdot \left(\frac{1}{4}\right)^{k-1}.$$

Combining the results from the first two parts, we obtain:

$$\begin{aligned} P(E \cup F) &= \underbrace{\left(\frac{1}{4}\right)^{k-1}}_{P(E)} + \underbrace{1 - \left(\frac{1}{2}\right)^k}_{P(F)} - \underbrace{\frac{1}{2} \cdot \left(\frac{1}{4}\right)^{k-1}}_{P(E \cap F)} \\ &= \frac{1}{2} \cdot \left(\frac{1}{4}\right)^{k-1} + 1 - \left(\frac{1}{2}\right)^k. \end{aligned}$$

Exercise 2. Let $(\Omega, \mathcal{A}, P)$ be a probability space, $B \in \mathcal{A}$ with $P(B) > 0$ and I a countable index set. Show the following statements:

(i) The map

$$\tilde{P} : \mathcal{A} \rightarrow [0, 1], \quad A \mapsto \tilde{P}(A) := P(A \mid B)$$

is a probability measure on $(\Omega, \mathcal{A})$. The probability space $(B, \mathcal{A}|_B, \tilde{P}|_B)$ is called *trace of $(\Omega, \mathcal{A}, P)$ on B* .

(ii) Let $\{B_i \mid i \in I\} \subseteq \mathcal{A}$ be a measurable *partition* of B , i.e., $\bigcup_{i \in I} B_i = B$ and $B_i \cap B_j = \emptyset$ for all $i, j \in I$ with $i \neq j$. For all $A \in \mathcal{A}$ we then have the *law of total probability*

$$P(A \cap B) = \sum_{i \in I} P(A \mid B_i)P(B_i).$$

(iii) For each $A \in \mathcal{A}$ with $P(A) > 0$ and every measurable partition $\{E_i \mid i \in I\}$ of Ω with $P(E_i) > 0$ for all $i \in I$, we have the *Bayes rule*

$$P(E_i \mid A) = \frac{P(A \mid E_i)P(E_i)}{\sum_{j \in I} P(A \mid E_j)P(E_j)} \quad \text{for all } i \in I.$$

Solution:

(i) A map $\mu : \mathcal{A} \rightarrow [0, \infty]$ is called a probability measure, if

$$(1) \quad \mu(\emptyset) = 0,$$

$$(2) \quad \forall A \in \mathcal{A} : \mu(A) \geq 0,$$

$$(3) \quad \mu\left(\bigcup_{i \in \mathbb{N}} A_i\right) = \sum_{i \in \mathbb{N}} \mu(A_i) \text{ for all countable collections } (A_i)_{i \in \mathbb{N}} \text{ of pairwise disjoint sets in } \mathcal{A},$$

and

$$(4) \quad P(\Omega) = 1.$$

We will verify these properties for $\tilde{P}$:

(1) Probability of the empty set:

$$\tilde{P}(\emptyset) \stackrel{\text{def.}}{=} P(\emptyset \mid B) = \frac{P(\emptyset \cap B)}{P(B)} = \frac{P(\emptyset)}{P(B)} \stackrel{(*)}{=} 0,$$

where we're allowed to make the final transformation $(*)$ since $P(B) > 0$ as per problem statement, and $P(\emptyset) = 0$ due to the fact that P itself is a probability measure.

(2) Non-negativity:

$$\forall A \in \mathcal{A} : \tilde{P}(A) = P(A \mid B) = \frac{\overbrace{P(A \cap B)}^{\geq 0 \text{ (**)}}}{\underbrace{P(B)}_{> 0}} \geq 0.$$

(**) $A, B \in \mathcal{A} \implies (A \cap B) \in \mathcal{A} \implies P(A \cap B) \geq 0$ since, again, P itself is a probability measure.

(3) σ -additivity:

Let $(A_n)_{n \in \mathbb{N}}$ be a collection of pairwise disjoint ($A_i \cap A_j = \emptyset \ \forall i \neq j$) sets in $\mathcal{A}$. Then:

$$\tilde{P} \left(\bigcup_{n \in \mathbb{N}} A_n \right) = P \left(\bigcup_{n \in \mathbb{N}} A_n \middle| B \right) = \frac{P \left(\left(\bigcup_{n \in \mathbb{N}} A_n \right) \cap B \right)}{P(B)} = \frac{P \left(\bigcup_{n \in \mathbb{N}} (A_n \cap B) \right)}{P(B)}.$$

Since the A_n are disjoint, the sets $A_n \cap B$ are disjoint as well. Thus, we can use the σ -additivity of the measure P to obtain:

$$P \left(\bigcup_{n \in \mathbb{N}} (A_n \cap B) \right) = \sum_{n \in \mathbb{N}} P(A_n \cap B).$$

Putting everything together, we get:

$$\tilde{P} \left(\bigcup_{n \in \mathbb{N}} A_n \right) = \frac{\sum_{n \in \mathbb{N}} P(A_n \cap B)}{P(B)} = \sum_{n \in \mathbb{N}} \frac{P(A_n \cap B)}{P(B)} = \sum_{n \in \mathbb{N}} P(A_n | B) = \sum_{n \in \mathbb{N}} \tilde{P}(A_n).$$

(4) Normalization:

Finally, we compute

$$\tilde{P}(\Omega) = P(\Omega | B) = \frac{P(\Omega \cap B)}{P(B)} = \frac{P(B)}{P(B)} \stackrel{(***)}{=} 1.$$

Again, we use $P(B) > 0$.

□

(ii) Using the definition of conditional probability, we can rewrite the right-hand side of the equation as:

$$\sum_{i \in I} P(A | B_i) P(B_i) = \sum_{i \in I} \frac{P(A \cap B_i)}{P(B_i)} P(B_i) = \sum_{i \in I} P(A \cap B_i).$$

Since the B_i are disjoint, the sets $A \cap B_i$ are pairwise disjoint. Thus, we can use the additivity of the measure P to obtain:

$$\sum_{i \in I} P(A \cap B_i) = P \left(\bigcup_{i \in I} (A \cap B_i) \right) = P \left(A \cap \bigcup_{i \in I} B_i \right) = P(A \cap B) \quad \square$$

(iii) Again, we begin with the definition of conditional probability as

$$P(E_i | A) = \frac{P(E_i \cap A)}{P(A)}.$$

For the numerator, we have

$$P(E_i \cap A) = P(A | E_i) P(E_i) (= P(E_i | A) P(A)).$$

And we can rewrite $P(A)$ using the law of total probability as:

$$P(A) \stackrel{(*)}{=} P(A \cap \Omega) = \sum_{i \in I} P(A | E_i) P(E_i).$$

(*) Remember that the E_i build a partitioning of $\Omega \supseteq A$. Putting everything together, we get:

$$P(E_i | A) = \frac{P(A | E_i) P(E_i)}{\sum_{j \in I} P(A | E_j) P(E_j)}.$$

Exercise 3. For each of the following settings provide all modeling details and rigorous computations for your results.

- (i) You play Russian roulette with a six shooter revolver and there are precisely two bullets in neighboring chambers. Your opponent goes first, spins the barrel, pulls the trigger and the gun does not fire. Now it is your turn. Should you spin the barrel again or pull the trigger right away? What is the probability of survival in each case?
- (ii) I have two cards in a closed opaque box, one is red on both sides and the other is red on one and blue on the other side. I pull out one card and place it on the table and a red side faces up. What's the probability that the other side is also red?

Solution:

- (i) **“General construction principle:”** To model a discrete experiment $\Omega = \prod_{i \in [n]} \Omega_i$ comprised of a sequence of n part-experiments Ω_i , one assumes an underlying/unknown probability space $(\tilde{\Omega}, \tilde{\mathcal{A}}, P)$ and uses $(\Omega_i, \mathcal{A}_i)$ -valued RVs $X_i : \tilde{\Omega} \rightarrow \Omega_i$ (for $i \in [n]$) to model the outcome of the i -th part-experiment (Ω_i also assumed to be discrete). We can now build a probability measure on $(\Omega, \mathcal{A} = \mathcal{P}(\Omega))$, i.e., a probability measure over sequences of n outcomes in the different Ω_i , iteratively as follows:

The first random variable X_1 induces a probability measure P_{X_1} on $(\Omega_1, \mathcal{A}_1)$. This is simply the probability measure modeling a single (the first) experiment and we can characterize it fully via the pmf $P_{X_1}(\{\omega_1\})$ for all $\omega_1 \in \Omega_1$. For each outcome in Ω_1 , we can then ask, what the probability measure over $(\Omega_2, \mathcal{A}_2)$, the second experiment, should be, given what we have observed in the first one. Mathematically, this is fully characterized via the conditional pmf $P_{X_2|X_1=\omega_1}(\{\omega_2\})$, i.e., “the probability of getting ω_2 in the second experiment, given that the first experiment gave ω_1 .” We can now proceed iteratively with the following experiments.

Overall, one can then show that P_{X_1} together with $P_{X_k|X_1=\omega_1, \dots, X_{k-1}=\omega_{k-1}}$ for all $k \in \{2, \dots, n\}$ uniquely induces a probability measure on the space of sequences $(\Omega, \mathcal{A} = \mathcal{P}(\Omega))$ via

$$P(\{\omega\}) = P(\{\omega_1, \dots, \omega_n\}) := P_{X_1}(\{\omega_1\})P_{X_2|X_1=\omega_1}(\{\omega_2\}) \dots P_{X_n|X_1=\omega_1, \dots, X_{n-1}=\omega_{n-1}}(\{\omega_n\}).$$

With this, we can now model our experiment as a “two-round” experiment with $\Omega = \Omega_1 \times \Omega_2$, where $\Omega_1 = \Omega_2 = \{e, ne\}$ (e = “empty”, ne = “not empty”), and X_i modelling the i -th time pulling the trigger. In the beginning, we spin the chamber, which hints at a uniform distribution over the six chambers initially. Since 2 chambers are filled and 4 are empty, we have

$$P_{X_1}(e) = \frac{4}{6}, \quad P_{X_1}(ne) = \frac{2}{6}.$$

According to the exercise the first shot did not fire.

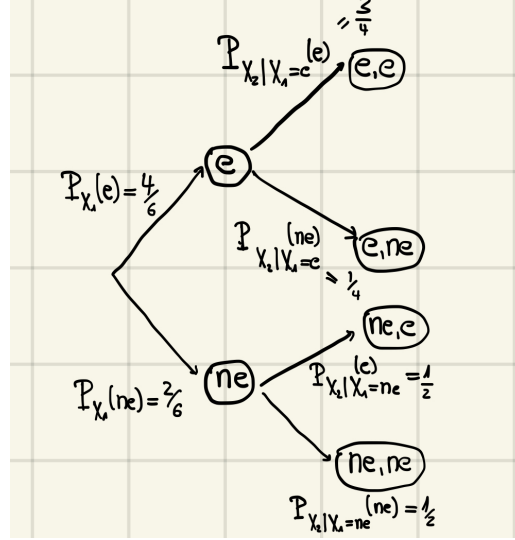
First, let us assume you spin the chamber again. Then we have $P_{X_2|X_1=e} = P_{X_1}$, which we can also write as $P_{X_2|X_1=e}(\text{“survival”}) = P_{X_1}(\{e\}) = \frac{2}{3}$.

Second, assume you directly pull the trigger again. In this case, note that pure “symbol manipulation” does not tell us, what $P_{X_2|X_1=e}$ is. Since we have only modeled “empty” and “not-empty” as events, we need to reason about the situation to determine, which events are how likely after observing an “empty” event in the first round. (Instead, we could have also indexed the chamber slots and modeled the index that came up when pulling the trigger and then reasoning from there, whether we would have died or not.) In this case, we need to actually prescribe $P_{X_2|X_1=e}$ from the description of the situation: We know that there are four possibilities of having hit an empty chamber, each of which was equally likely. Since the empty chambers are all “in one row”

and the chamber moves forward in a fixed direction, only a single one of them is followed by a bullet, whereas 3 of them are followed by another empty chamber.

$$P_{X_2|X_1=e}(\{e\}) = \frac{3}{4}.$$

This can be also illustrated as follows



Hence, the probability of survival is $\frac{3}{4}$ when pulling the trigger directly compared to $\frac{2}{3}$ when spinning again.

Note again how an understanding of the experiment mechanics, i.e., how revolvers work (moving by one slot at a time in one direction when pulling the trigger) as well as a “leap of faith assumption” (translating “spinning” to “uniform distribution”) where paramount in writing down the probability measures we chose and thus in arriving at the final results. These interpretations and choices are precisely the process of “modeling.”

For example, you could have decided that spinning the barrel does not result in a uniform distribution since bullets have weight so may end up at the bottom with a higher probability (maybe 3%?). Maybe you are happy to assume that the average person loads the bullets into the two right-pointing chambers (with what probability?). Together with some measurements (or assumptions) of how much force you and your opponent apply in a “typical revolver spin,” you could then build a more refined choice for P_{X_1} and $P_{X_2|X_1=e}$.

- (ii) Assume an underlying probability space $(\Omega, \mathcal{A}, P)$ and two random variables $X_{\text{card}} : \Omega \rightarrow \{1, 2\}$ indicating the drawn card (we assume $P_{X_{\text{card}}} = \text{Lap}_{\{1,2\}}$) and $X_{\text{col}} : \Omega \rightarrow \{\text{“red”}, \text{“blue”}\}$ indicating the visible “turned-up” color on the table.

With Bayes’ theorem we then obtain

$$\begin{aligned}
 & P(X_{\text{card}} = 1 \mid X_{\text{col}} = \text{“red”}) \\
 &= \frac{P(X_{\text{card}} = 1) P(X_{\text{col}} = \text{“red”} \mid X_{\text{card}} = 1)}{P(X_{\text{card}} = 1) P(X_{\text{col}} = \text{“red”} \mid X_{\text{card}} = 1) + P(X_{\text{card}} = 2) P(X_{\text{col}} = \text{“red”} \mid X_{\text{card}} = 2)} \\
 &= \frac{\frac{1}{2} \cdot 1}{\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2}} = \frac{2}{3}.
 \end{aligned}$$

So indeed, the probability that the backside is also red is $\frac{2}{3}$. Intuitively, there are four sides that could be facing up before we get any information. By the assumptions, each of them is equally

likely. The fact that we obtain the “evidence” that we see a red side facing up, eliminates the possibility of seeing the blue side up. This leaves us with three options, either side of the red-red card or the red side of the red-blue card. Among these three (equally likely options), the other side is also red in two of them.

Exercise 4. We uniformly choose a real number from $[0, 1]$, square it, and represent the result by a real-valued random variable X .

- (i) What is the cumulative distribution function of X ?
- (ii) What is the probability density function of X ?
- (iii) What is the probability that $X > \frac{1}{4}$. Try to find an answer in your head intuitively first.

Solution: Let U be the random variable uniformly distributed on $[0, 1]$, i.e., $U \sim \text{Unif}(0, 1)$. We then have $X = U^2$.

- (i) As we have

$$F_X(x) = P(X \leq x) = P(U^2 \leq x) = P(U \leq \sqrt{x})$$

we obtain the cdf as

$$F_X(x) = P(U \leq \sqrt{x}) = \begin{cases} 0 & \text{if } x < 0, \\ \sqrt{x} & \text{if } 0 \leq x \leq 1, \\ 1 & \text{if } x > 1. \end{cases}$$

- (ii) The probability density function is then given by

$$f_X(x) = \frac{\partial}{\partial x} F_X(x) = \begin{cases} \frac{1}{2\sqrt{x}} & \text{if } 0 < x \leq 1, \\ 0 & \text{if otherwise.} \end{cases}$$

- (iii) We compute

$$P(X > \frac{1}{4}) = 1 - F_X(\frac{1}{4}) = 1 - \sqrt{\frac{1}{4}} = \frac{1}{2}.$$